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PROJECT NAME	PocketSmart AI: Your Smart Budget & Recommendation Assistant

Phase 06: Project Testing

1. Introduction

This phase validates the reliability, accuracy and contextual awareness of the PocketSmart AI recommendation engine. The system was tested with real-world scenarios across Home Interiors, Party Planning and Jewelry Suggestions. Because the output is produced by Gemini 2.5, testing concentrates on structure, budget adherence, relevance and error handling rather than exact wording.

1.1 Test Strategy

Test type	Goal	Method
Functional	Every route and form works end to end	Manual test cases through the interface
AI quality	Relevant items, INR prices, correct platforms	Review each response against a checklist
Budget adherence	Item costs do not exceed the budget	Add item costs and compare with remaining budget
Validation	Invalid input is handled gracefully	Empty, negative, very large and non-numeric values
Security	Protected pages need login	Open pages while logged out; test expired sessions
Usability	Forms are readable on phone and desktop	Resize the browser; test on a phone
Performance	Response time is acceptable	Time 10 runs per planner

1.2 Test Environment

- Windows, Python 3.10 or newer, Chrome browser.
- Application running locally at port 8000.
- Valid Gemini 2.5 API key on the free tier.
- A newly registered test user for each test round.

2. Functional Test Cases

ID	Scenario	Input / Steps	Expected result	Status
TC-01	Register new user	Valid username, email, matching passwords	Account created; redirect to login	Pass
TC-02	Register with mismatch	Passwords differ	Error message; no account	Pass
TC-03	Login valid	Correct credentials	Token issued; dashboard shown	Pass
TC-04	Login invalid	Wrong password	Incorrect username or password message	Pass
TC-05	Protected page	Open dashboard while logged out	Access denied or redirect	Pass
TC-06	Home plan	Budget 5000, 5 lights, 4 fans, 2 furniture, 1 dining table, Living Room and Kitchen	Category plan with links for five platforms	Pass

ID	Scenario	Input / Steps	Expected result	Status
TC-07	Party plan	Budget 5000, 3 guests, wedding, home venue, catering and entertainment	Catering, entertainment and contingency with relevant links	Pass
TC-08	Jewelry, text only	Budget 5000, occasion Birthday	Three or more items, tips, seven platform links	Pass
TC-09	Jewelry with image	Same input plus outfit photo	Outfit analysis with colours, style, formality	Pass
TC-10	History	Open history after three plans	Three cards, newest first; details open	Pass
TC-11	Logout	Click Logout	Token blacklisted; redirect to login	Pass
TC-12	Session expiry	No activity for over 30 minutes	Session removed by cleanup task	Pass

3. AI Output Evaluation

Gemini 2.5 responses were evaluated for quality, budget adherence and platform accuracy.

Planner	Budget	Allocated	Remaining	Observation
Home	5000	4500	500	Lighting 1500, ceiling fans 2000, furniture 1000; links for Amazon, Flipkart, IKEA, Myntra and Ajio
Party	5000	5000	0	Catering 2000, entertainment 2000, contingency 1000; home used as venue at zero cost
Jewelry	5000	4200	800	Bracelet 500, ring 700, watch 3000; minimal casual style matched a blue and white outfit

4. Defects, Optimization and Conclusion

4.1 Observations and Recommended Fixes

ID	Observation	Likely cause	Recommended fix
OBS-01	Home and Jewelry forms show a dollar label while results use INR	Template label not updated	Change labels to INR
OBS-02	Party venue card shows undefined location	Model did not return location fields	Show Not specified when a field is missing
OBS-03	Home item prices do not add up to category allocation	Model estimates loosely	Recalculate totals from item prices in the backend
OBS-04	Extra text around the AI reply can break parsing	Model adds explanations	JSON extraction helper removes surrounding text
OBS-05	Category names appear as Ceiling_fans	Raw key displayed	Replace underscores and capitalize
OBS-06	Non-image file accepted as outfit	No file validation	Accept only image types within a size limit

4.2 Optimization

- Prompts tightened with market (India), INR, fixed JSON structure and a strict budget rule.
- Input validation added for budgets and counts.
- Fallback recommendations and a clear message when the AI returns insufficient results.
- Category-specific platform mapping to avoid irrelevant links.

4.3 Conclusion

All twelve core test cases passed. The observations in section 4.1 are minor display and accuracy issues recorded for improvement; they do not block the core functionality. The system returned structured, budget-aware plans for each planner in roughly 5 to 15 seconds and is ready for demonstration.